



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PURWANCHAL CAMPUS,
DHARAN-8**

**A MINOR PROJECT PROPOSAL
ON
NEPALI LIFESTYLE BASED DIET AND FITNESS ADVISORY
SYSTEM**

SUBMITTED BY:

**HIMAL BHANDARI (PUR080BCT034)
KSHITIZ PORTEL (PUR080BCT039)
MANAV ACHARYA (PUR080BCT041)
MAUSAM PARAJULI (PUR080BCT045)**

SUBMITTED TO:

**DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
IOE - PURWANCHAL CAMPUS, DHARAN-8**

SUBMISSION DATE:

2083 / 02 / 27

TABLE OF CONTENTS

1	Introduction	1
1.1	Background Study	1
1.2	Problem Definition	1
1.3	Objectives	1
1.4	Scope of Project	1
2	Literature Review	2
2.1	Association of Lifestyle and Food with Health	2
2.2	Mobile App for Management and Treatment of Gestational Diabetes in Nepal	2
2.3	AI-Driven Personalized Nutrition Systems	2
2.4	AI-Enabled Health Monitoring Applications	2
2.5	SmartCal: Computer Vision-Based Calorie Estimation	3
2.6	Gaps and Challenges	3
3	Methodology	4
3.1	Proposed Block Diagram	4
3.2	Tools and Techniques	4
3.2.1	Tools	4
3.2.2	Techniques	5
3.3	Data Sources	5
3.4	Implementation	6
3.4.1	Food Detection and Segmentation	6
3.4.2	Nutritional Information Retrieval Using Cosine Similarity	6
4	Expected Output and Schedule	7
4.1	Expected Output	7
4.2	Expected Schedule	7
	References	8

1 Introduction

1.1 Background Study

A diet and fitness advisory system is a personalized recommendation platform that processes individual user data such as age, weight, activity level, medical conditions, social constraints and so on to deliver customized health guidance. Globally, lifestyle-related diseases such as obesity, diabetes, and cardiovascular disorders are rising at an alarming rate, largely driven by poor dietary choices and physical inactivity, creating a strong demand for intelligent health advisory tools.

In Nepal, rapid urbanization and the growing influence of processed food culture have gradually replaced traditional eating habits, leading to widespread nutritional imbalances. Despite this, most available health platforms are built around Western dietary norms and offer no consideration for Nepali food culture, local ingredients, or regional eating patterns, making them largely ineffective for Nepali users.

In this context, the *Nepali Lifestyle-Based Diet and Fitness Advisory System* is designed to provide customized guidance tailored to user profiles and local dietary patterns.

1.2 Problem Definition

Existing health applications provide generic recommendations that overlook individual health conditions and local dietary habits. Since traditional Nepali food patterns are rarely reflected in such platforms, there is a clear need for a system that processes user-specific data and delivers personalized, culturally relevant diet and fitness guidance.

1.3 Objectives

- To build an automated application that tracks daily user health metrics and provides personalized, easy-to-understand lifestyle coaching.
- To automate food tracking by combining YOLO polygon masking for photo segmentation with vector similarity matching for local names.

1.4 Scope of Project

The project focuses on developing a web-based advisory system that collects user health-related information, including age and dietary preferences, and generates personalized diet and fitness recommendations aligned with Nepali food culture. The system includes user profile management, food tracking, recommendation generation, and health reporting features. The system is intended for wellness purposes only and does not extend to medical diagnosis or clinical treatment.

2 Literature Review

2.1 Association of Lifestyle and Food with Health

Chaudhary et al. [1] investigated the association between lifestyle, food, and bone mineral density (BMD) among adults aged 50 years and above in Kathmandu, finding that dietary habits significantly impacted bone health due to a systemic lack of calcium and other vital minerals in local food consumption.

Similarly, Sharma et al. [2] evaluated the nutritional contents of commercially produced complementary foods (CPCFs) marketed for young children in Nepal. Their research analyzed 61 unique available CPCFs and revealed that none met the absolute energy and nutrient composition thresholds set by the WHO SEAR NPPM, with more than half exceeding the maximum thresholds allowed for sugar and sodium. Together, these studies highlight a critical lack of proper nutritional optimization in locally available foods and underscore their direct impact on public health.

2.2 Mobile App for Management and Treatment of Gestational Diabetes in Nepal

Shanmugavel et al. [3] developed a mobile application to monitor and manage Gestational Diabetes Mellitus (GDM) at Dhulikhel Hospital, specifically tailored for low-income Nepali users. The system successfully served the highly specific purpose of tracking GDM in users and helping them improve their health outcomes. However, a major limitation of this application is its narrow use case, as it is restricted exclusively to patients with GDM. Furthermore, the system relies heavily on human medical experts to consult with users and manually input data into the platform.

2.3 AI-Driven Personalized Nutrition Systems

Tsolakidis et al. [4] reviewed numerous artificial intelligence and machine learning-driven technologies for personalized nutrition, categorizing them into nutrition, recipe, and restaurant recommendation systems. Their research demonstrated that such technologies are highly effective at guiding individuals toward healthier lifestyles. These systems move beyond primitive calorie counting by providing tailored dietary advice based on individual health conditions and nutritional goals. However, a common gap among these systems is that they do not account for the cultural and dietary restrictions unique to diverse groups in Nepal, such as specific fasting (*vrata*) and mourning rituals where individuals are strictly limited to certain food groups.

2.4 AI-Enabled Health Monitoring Applications

Bhatta [5] discussed AI applications in diabetes care, highlighting the utility of mobile applications, wearable devices, dietary planning algorithms, and predictive analytics. The study emphasized that user-centered design, remote monitoring, and personalized dietary recommendations are key pillars for successful digital healthcare systems. The paper also identified critical challenges, including data quality, privacy concerns, and the necessity of clinical integration. These architectural considerations are highly relevant when designing any AI-assisted dietary monitoring platform.

2.5 SmartCal: Computer Vision-Based Calorie Estimation

The *SmartCal* system, developed by Manandhar et al. [6], uses a Mask R-CNN model trained on images of common Nepali food items such as rice, lentils, spinach, and eggs. Food images are captured via an ESP32-CAM module and processed through an instance segmentation network. While Mask R-CNN provides precise pixel-level identification of food objects and achieves high detection accuracy, the underlying model requires significant computational resources and exhibits longer inference times. This makes it less viable for real-time deployment on lightweight mobile or web-based applications.

2.6 Gaps and Challenges

A critical evaluation of the existing literature reveals that most nutrition recommendation systems are developed using international datasets, completely overlooking culturally specific dietary restrictions common in Nepal, such as religious fasting and localized eating habits. Similarly, existing local food recognition systems are either limited in dataset scope or rely on computationally expensive segmentation architectures that are poorly suited for real-time application.

To bridge these gaps, our proposed system aims to be lightweight and computationally efficient by leveraging the YOLO (You Only Look Once) family of object detection and segmentation algorithms. By training specifically on a custom Nepali cuisine dataset, the system will provide real-time inference and deliver personalized recommendations that actively respect local dietary practices, medical profiles, and cultural restrictions.

3 Methodology

3.1 Proposed Block Diagram

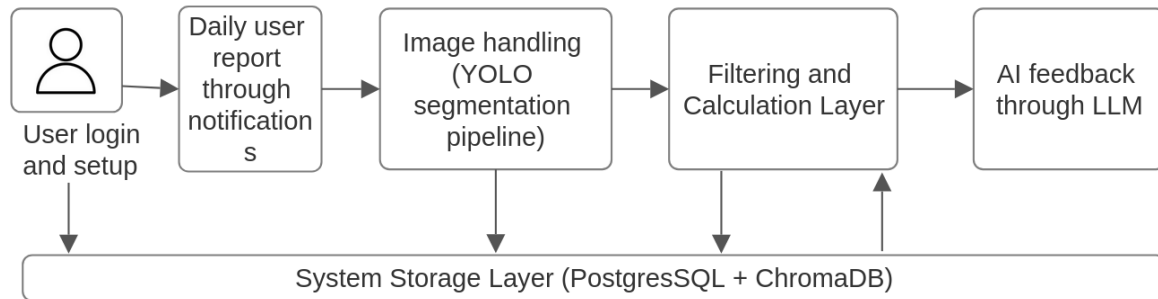


Fig: Proposed System Architecture and Data Processing Workflow

3.2 Tools and Techniques

3.2.1 Tools

- **Developmental Libraries and Frameworks**

- **React:** Used for building a dynamic, responsive, and intuitive user interface for the frontend application.
- **Django:** Utilized as the primary backend framework for robust user authentication, data management, and business logic implementation.
- **FastAPI Framework:** Employed to build high-performance, asynchronous APIs specifically for handling machine learning inferences and heavy computational tasks.

- **Artificial Intelligence and Computer Vision**

- **Ultralytics YOLO:** Used for real-time computer vision tasks, specifically instance segmentation to detect and mask food items from images.
- **PyTorch / TensorFlow:** Chosen as the foundational deep learning frameworks for training, fine-tuning, and running the AI models.

- **Databases**

- **PostgreSQL:** Used as the primary relational database management system (RDBMS) to securely store structured user profiles, health logs, and nutritional data.
- **ChromaDB:** Utilized as the vector database to store and efficiently query high-dimensional embeddings for local food name matching.

- **Data Pipeline, Validation, and Orchestration**

- **LangChain Orchestrator:** Used to manage the workflow pipeline between the user inputs, database retrievals, and Large Language Models (LLMs).

- **Pydantic Validation Engine:** Employed to enforce strict data parsing and type validation across the backend APIs and LLM outputs.

3.2.2 Techniques

1. **Setting Up Profiles:** Gathers basic user information like age, weight, health conditions, and social constraints while connecting to a standard food database that holds nutritional facts for local items.
2. **Finding Food in Photos:** Takes a top-down picture of a plate and processes it through an AI model. The system uses YOLO to trace a custom polygon mask around each food item to pinpoint its exact shape and separate it from other foods on the plate.
3. **Matching Local Names:** Converts the raw words identified by the camera and user's input into digital vector points. It then measures the cosine similarity/distance between these points in a vector database to match local everyday food names (like "bhat") to their official database entries (like "White Rice").
4. **Math and Health Screening:** Uses standard code in the backend to calculate the total nutrients based on the food portions. At the same time, it checks the user's stored health profile to automatically filter out or flag foods that conflict with their medical conditions and social constraints.
5. **Clean AI Responses:** Puts the calculated numbers into a strict, pre-defined template before sending it to a Large Language Model. This forces the LLM to reply using a fixed data structure (e.g., JSON) so it does not hallucinate wrong calculations or disrupt the application layout.
6. **Local Testing and Hosting:** Checking individual code components and mathematical calculations to ensure they work smoothly without errors. Once verified, the frontend can be deployed using Vercel, and the backend via platforms like Render or Railway. Alternatively, the system can run completely locally by utilizing a quantized local LLM for inference.

3.3 Data Sources

Our system asks the user about a dozen one-time setup questions initially, which are used to set feasible and healthy dietary goals. Regarding the local nutrition database, we have bifurcated and compiled various reference datasets to create a comprehensive master database of our own. Some of the major data sources include:

- *Nepal Food Composition* by Santosh Khatiwada
- *Nepali Meals Dataset* by AMITPANT7
- *Nepal Nutrition and Food Security Portal*

3.4 Implementation

This proposed system utilizes a combination of computer vision algorithms, semantic vector spaces, and mathematical optimization models to execute the core features of the fitness and dietary advisory platform.

3.4.1 Food Detection and Segmentation

The system uses a YOLO-based instance segmentation model to automatically detect and delineate food items from meal images. A custom dataset of common Nepali dishes is curated and annotated with precise polygon masks to train the architecture.

During the inference phase, a top-down image uploaded by the user is processed through the trained convolutional network. The network predicts bounding boxes, class probabilities, and corresponding pixel-level masks. Overlapping or redundant bounding boxes for the same food item are filtered out using Non-Maximum Suppression (NMS), leaving precise, clean food regions for subsequent spatial and quantitative analysis. These segmented pixel regions are then mapped against the user's calibrated household baseline templates to accurately estimate food portions and weight variables.

3.4.2 Nutritional Information Retrieval Using Cosine Similarity

The text labels generated by the computer vision pipeline might not identically match the exact string keys stored in the nutritional master database. For instance, colloquial variations like "bhat", "rice", and "white rice" represent the same fundamental food entity. To handle this lexical gap, the system implements a semantic similarity search using high-dimensional vector embeddings and a ChromaDB vector database.

Detected food labels are mapped into a dense vector space using a pre-trained text embedding model. Concurrently, all standard food records stored in our nutritional database are vectorized using the same model and indexed within ChromaDB. The semantic alignment between the user's detected item vector and the database entry vector is quantified using Cosine Similarity, which computes the cosine of the angle between two non-zero vectors in an inner product space:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

where:

- A = Embedding vector of the detected food label text.
- B = Embedding vector of a candidate database food entry.

The system selects the database record that yields the highest cosine similarity score above a predefined confidence threshold.

4 Expected Output and Schedule

4.1 Expected Output

This system is expected to recognize and segment local Nepali food items from photos using Ultralytics YOLO computer vision pipeline and calculate exact nutritional contents by matching the names of cuisine using a dual-database layer. Then, a structured LLM-backed engine is expected to generate personalized dietary feedback while following the constraints unique to Nepal like religious fasting or medical profiles. The system is expected to achieve a minimum of 80% mean Average Precision (mAP) for image-based food item segmentation and at least 85% accuracy for semantic name matching using vector space Cosine Similarity.

4.2 Expected Schedule

The planned timeline for the development, testing, and deployment of the project is mapped out in the Gantt chart below:

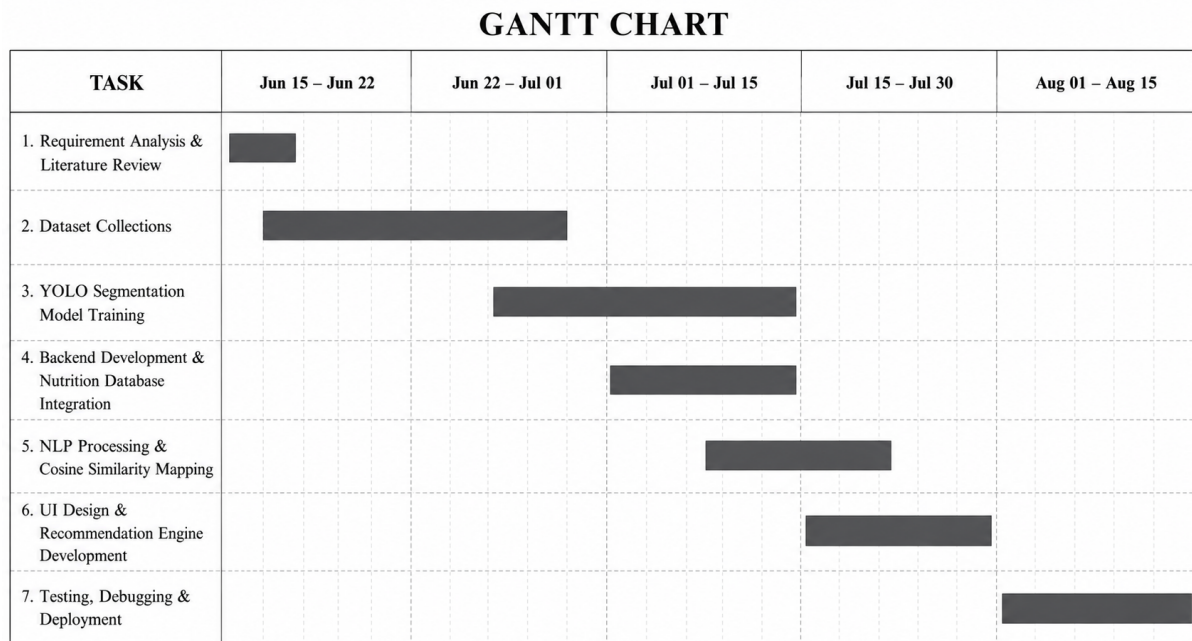


Figure 1: Gantt Chart Showing Project Timeline and Milestones

References

- [1] N. K. Chaudhary, M. N. Timilsena, D. R. Sunuwar, P. M. S. Pradhan, and R. K. Sangroula, "Association of lifestyle and food consumption with bone mineral density among people aged 50 years and above attending the hospitals of Kathmandu, Nepal," *Journal of Osteoporosis*, vol. 2019, no. 1, Art. no. 1536394, 2019.
- [2] N. Sharma, P. Lamichhane, P. Love, P. K. Poudel, and C. Bell, "Nutrient composition and health information on the labels of commercially produced complementary foods in Nepal," *Maternal & Child Nutrition*, vol. 22, no. 1, p. e70150, 2026.
- [3] A. Shanmugavel, P. R. Shakya, A. Shrestha, J. Nepal, A. Shrestha, J. F. Daneault, and S. Rawal, "Designing and developing a mobile app for management and treatment of gestational diabetes in Nepal: User-centered design study," *JMIR Formative Research*, vol. 8, no. 1, p. e50823, 2024.
- [4] D. Tsolakidis, L. P. Gymnopoulos, and K. Dimitropoulos, "Artificial intelligence and machine learning technologies for personalized nutrition: A review," *Informatics*, vol. 11, no. 3, p. 62, Aug. 2024.
- [5] R. P. Bhatta, "Opportunities and challenges of AI-based diabetes prediction: Global and regional perspectives," *Perspectives*, vol. 5, no. 1, pp. 107–113, 2026.
- [6] A. Manandhar, L. Subedy, C. K. Pandit, and P. Acharya, "SmartCal: Calorie estimation of local Nepali cuisine with deep learning-powered food detection," *KEC Journal of Science and Engineering*, vol. 9, no. 1, pp. 175–184, 2025.